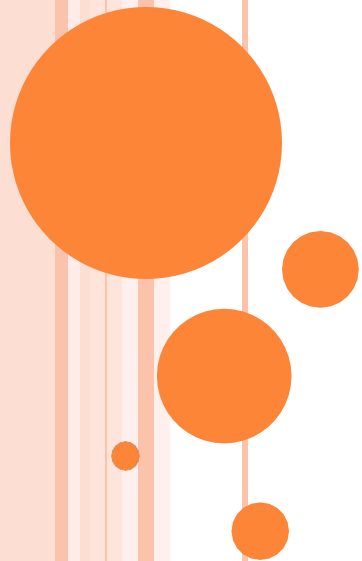


EC-201: Engineering Project Management

LECTURE NO 9: *Project Selection Criteria*



Broad

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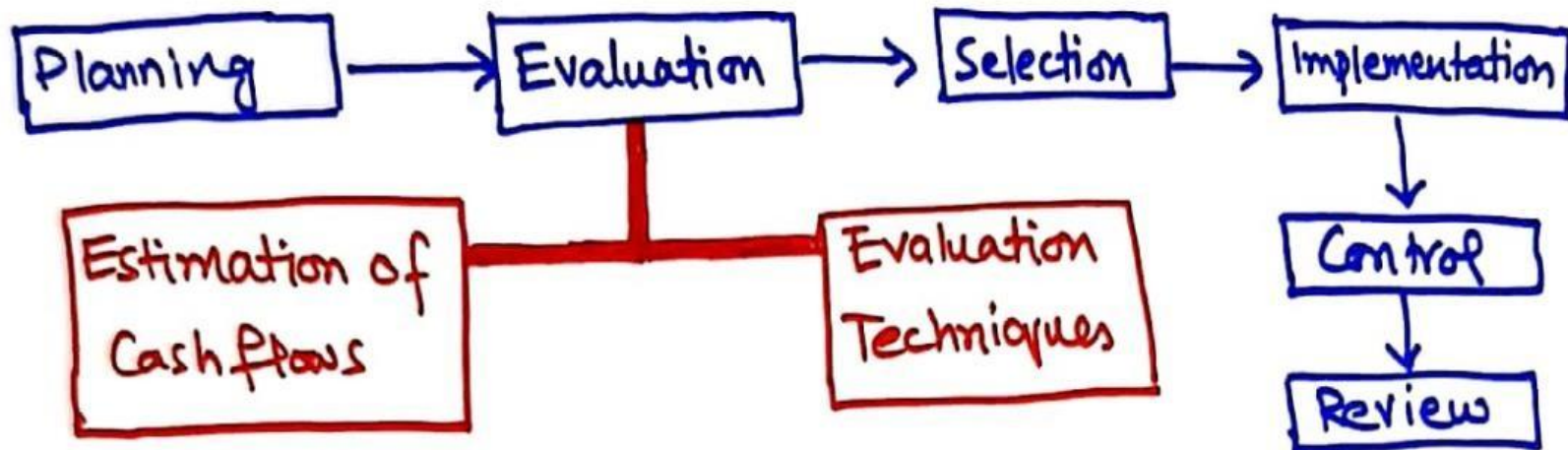
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9.1 Capital

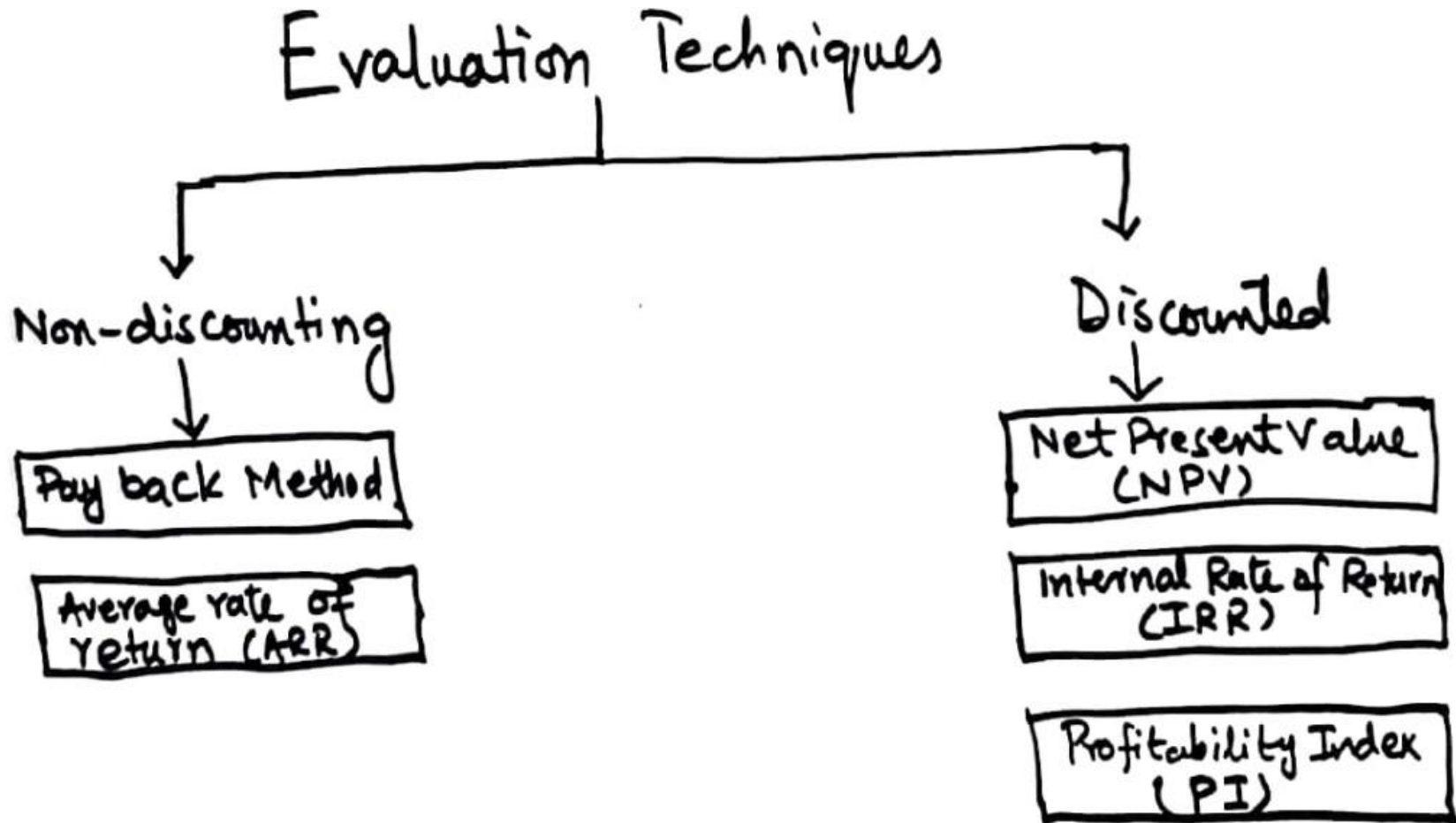
Capital Budgeting

↳ Capital Budgeting refers to the process of making decision regarding investment in fixed assets such as machinery, land, building etc.

Process of Capital Budgeting



9.2 Numeric Methods/Evaluation Techniques (1/38)



9.2 Numeric Methods/Evaluation Techniques (2/38)

- ↳ Now we will explain the concept of discounting as a time value of money with the help of an example:
If you have 100 rupees today in your pocket, then that is worth more than the same 100 rupees if you have that in a year later, In a year later the purchasing power of that 100 rupees will fall down.
- ↳ Another example can be: If you are receiving 1-lac rupee in future, the worth of that money in today's term might be 95000 rps. We will discuss this concept in detail in NPV

9.2 Numeric Methods/Evaluation Techniques (3/38)

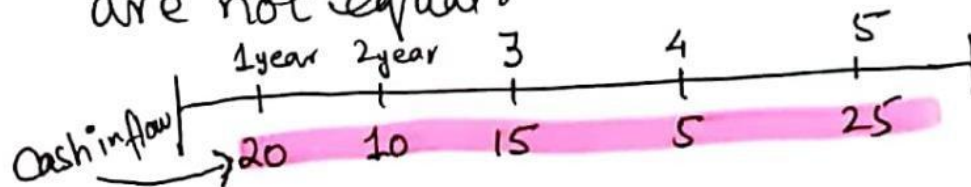
A. Non-discount techniques

* Pay Back Method

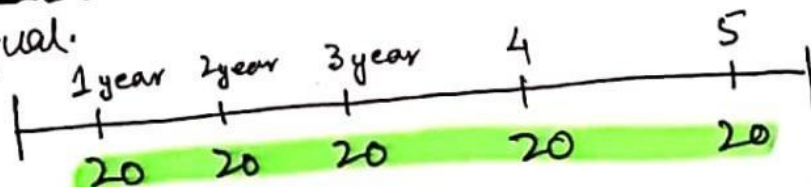
↳ How soon can we get our cash back?

$$\text{Pay back Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}} (\text{years})$$

↳ Cumulative Cash flows: When the cash inflows are not equal.



↳ Equal Cash flows: When the cash inflows are equal.



9.2 Numeric Methods/Evaluation Techniques (4/38)

A. Non-discount techniques

Example 1

Initial Investment = 25,00,000

Annual Cash Inflow = 5,00,000 for 8 years (Equal Cash Inflow)

Calculate Pay back Period.

Sol

$$\text{Pay back Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}} \text{ (years)}$$

$$= \frac{25,00,000}{5,00,000} \text{ (years)}$$

$\text{Pay back Period} = 5 \text{ years}$

9.2 Numeric Methods/Evaluation Techniques (5/38)

A. Non-discount techniques

Example 2

Suggest the Management using Pay back Period.

Machine A

- Initial Investment = 15,00,000
- Cash Inflow = 5,00,000

Solution

$$PBP_A = \frac{15,00,000}{5,00,000} \text{ (years)}$$

$$PBP_A = 3 \text{ years}$$

Go for Machine A

Machine B

- Initial Investment = 20,00,000
- Cash Inflow = 5,50,000

$$PBP_B = \frac{20,00,000}{5,50,000} \text{ (years)}$$

$$PBP_B = 3.6 \text{ years}$$

9.2 Numeric Methods/Evaluation Techniques (6/38)

A. Non-discount techniques

Example 3 (Related to Cumulative Cash flow)

An Industry is considering investment in a project which cost Rs. 6,00,000

Cash Inflows are Rs. 120,000, Rs. 140,000, Rs. 180,000, Rs. 2,00,000 & Rs. 250,000

Calculate Payback Period

9.2 Numeric Methods/Evaluation Techniques (7/38)

A. Non-discount techniques

Sol → 1st day of the 1st year

Years	Cash flows	Cumulative Cash flow
0	Initial Investment = 6,00,000 = -6,00,000 = (6,00,000)	(6,00,000) - <i>Cash outflow</i>
1	120,000	-600,000 + 120,000 = -480,000 = (480,000) - <i>Cash outflow</i>
2	140,000	-480,000 + 140,000 = -340,000 = (340,000)
3	180,000	-340,000 + 180,000 = -160,000 = (160,000)
4	200,000	-16000 + 200,000 = +40,000 → <i>Cash Inflow</i>
5	250,000	40,000 + 250,000 = 290000

→ end of 1st year
→ end of 4th year

9.2 Numeric Methods/Evaluation Techniques (8/38)

A. Non-discount techniques

- ↳ Between year 3 & 4, the initial investment of 6 lac (6,00,000) has been recovered.
- So how do we know exactly when (in years) it was recovered?
- ✓ It was recovered at the end of 3rd year, i.e. the 3rd year was over.

9.2 Numeric Methods/Evaluation Techniques (9/38)

A. Non-discount techniques

↳ Now payback period can be calculated as:

$$\text{PBP} = 3 \text{ years} + \frac{\text{Cumulative cash flow of 3}^{\text{rd}} \text{ year}}{\text{Cash flow of 4}^{\text{th}} \text{ year}}$$

$$= 3 \text{ years} + \frac{160,000}{200,000}$$

$$= 3 \text{ years} + 0.8$$

$$\boxed{\text{PBP} = 3.8 \text{ years}}$$

9.2 Numeric Methods/Evaluation Techniques (10/38)

A. Non-discount techniques

* Average Rate of Return (ARR)

↳ This method takes into account the earnings from the investment over its WHOLE LIFE

$$ARR = \frac{\text{Average Profit}}{\text{Average Investment}} \times 100$$

↳ Profit = Cash Inflows — whole depreciation — tax
✓ where whole depreciation = Initial Investment — Scrap value

$$\text{Average Profit} = \frac{\text{Profit}}{\text{No. of years}}$$

↳ Average Investment:

$$= \left[\frac{\text{Initial Investment} + \text{Scrap value}}{2} \right] + \text{Additional working Capital.}$$

9.2 Numeric Methods/Evaluation Techniques (11/38)

A. Non-discount techniques

Important Key Terms

- ✓ Depreciation: A reduction in the value of an asset over time due in particular to wear & tear
- ✓ Scrap value: The termination value of the project
- ✓ Additional working capital: Sometimes we need funds in between of the project

↳ The project with highest ARR will be selected

9.2 Numeric Methods/Evaluation Techniques (12/38)

A. Non-discount techniques

Example 4

- Initial Investment : 160,000

No Scrap Value

- Cash Inflows:

years	1	2	3	4	5
Rupees	56,000	48,000	30,000	64,000	80,000

- Depreciation on straight line basis (Assume a constant rate of depreciation)

- Tax Rate 40%

9.2 Numeric Methods/Evaluation Techniques (13/38)

A. Non-discount techniques

Sol

$$ARR = \frac{\text{Average Profit} \rightarrow ?}{\text{Average Investment} \rightarrow ?} \Rightarrow \textcircled{A}$$

$$\text{Profit} = [\text{Cash Inflows} - (\text{whole depreciation})] - \text{Tax} \Rightarrow \textcircled{B}$$

- Cash Inflows = 56,000 + 48,000 + 30,000 + 64,000 + 80,000

$$\text{Cash Inflows} = 2,78,000 \text{ rupees}$$

- Depreciation = Initial Investment - Scrap value

$$= 160,000 - 0$$

- Whole depreciation = 160,000 rupees

9.2 Numeric Methods/Evaluation Techniques (14/38)

A. Non-discount techniques

$$\begin{aligned} -\text{Tax} &= [\text{cash inflow} - \text{whole depreciation}] \times 40\% \\ &= [2,78,000 - 1,60,000] \times 40\% \\ &= (1,18,000) \times 40\% \end{aligned}$$

$$\text{Tax} = 47,200 \text{ rupees}$$

$$\textcircled{B} \rightarrow \text{Profit} = 2,78,000 - 1,60,000 - 47,200$$

$$\boxed{\text{Profit} = 70,800 \text{ rupees}}$$

$$\text{Average Profit} = \frac{\text{Profit}}{\text{No. of years}} = \frac{70,800}{5} = 14,160 \text{ rupees}$$

$$\boxed{\text{Average Profit} = 14,160 \text{ rupees}}$$

9.2 Numeric Methods/Evaluation Techniques (15/38)

A. Non-discount techniques

Average Investment:

$$= \left[\frac{\text{Initial Investment} + \text{Scrap Value}}{2} \right] + \text{Additional working capital}$$

$$= \left[\frac{160,000 + 0}{2} \right] + 0$$

$$\boxed{\text{Average Invest} = 80,000 \text{ rupees}}$$

$$\textcircled{A} \Rightarrow \text{ARR} = \frac{14,160}{80,000} \times 100$$

$$\boxed{\text{ARR} = 17.7\%}$$

9.2 Numeric Methods/Evaluation Techniques (16/38)

B. Discounted techniques

Important key concept

Q: How to calculate Discount factors (DFs)? when discounting tables are not provided.

e.g i) If Rate = 10%.

for; 1st year, D.F will be $(1 \div 1.10)^1$

2nd year, " " " $(1 \div 1.10)^2$

3rd year, " " " $(1 \div 1.10)^3$

ii) if Rate = 7%.

for; 1st year, D.F will be $(1 \div 1.07)^1$

2nd year, D.F will be $(1 \div 1.07)^2$

9.2 Numeric Methods/Evaluation Techniques (17/38)

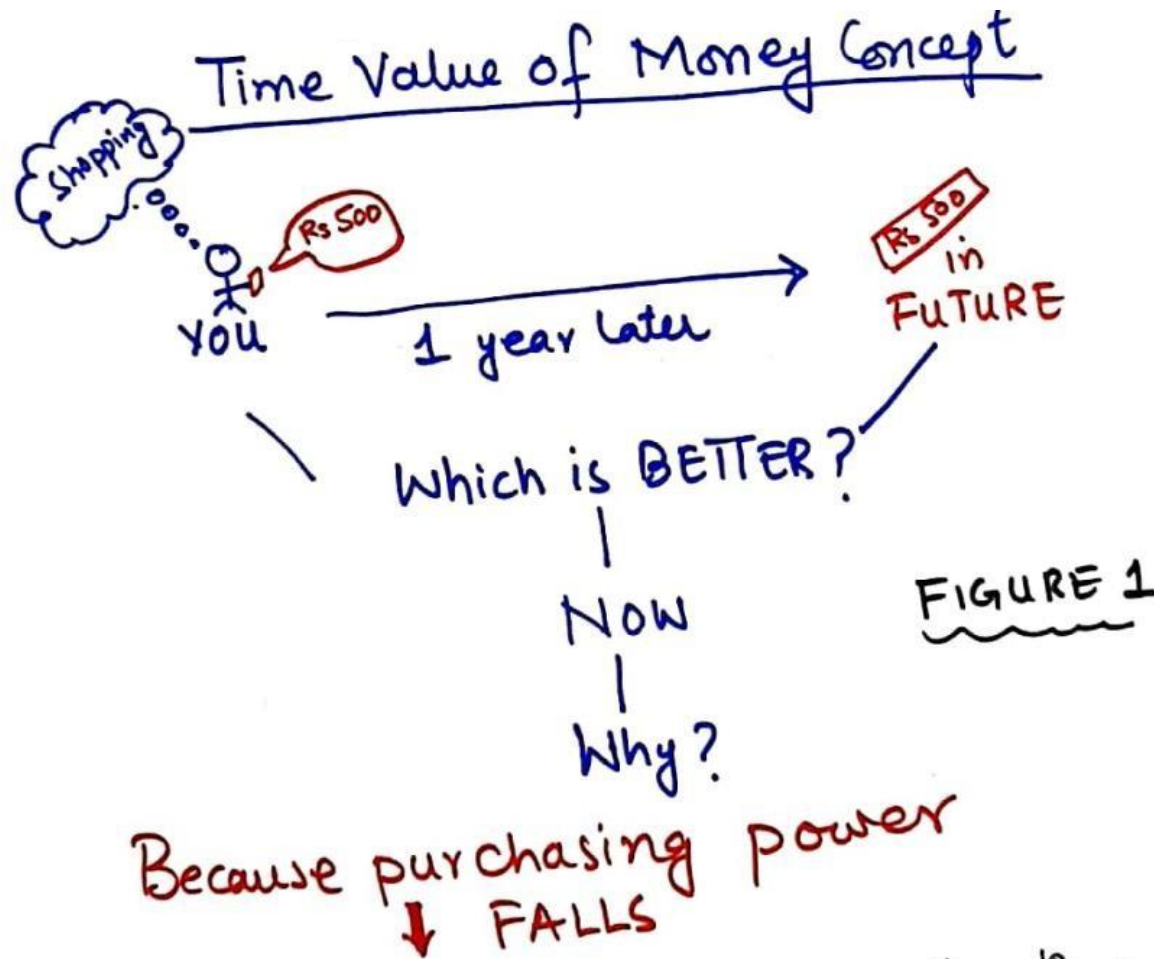
B. Discounted techniques

* Net Present Value (NPV)

- ↳ The NPV technique is a discounted cash flow method that considers time value of money in evaluating capital investments.
- ↳ It is a method of calculating the present value of cash flows (inflows + outflows) of an investment proposals using the cost of capital as an appropriate discounting rate.

9.2 Numeric Methods/Evaluation Techniques (18/38)

B. Discounted techniques



↳ Value of money Falls down as the time passes by

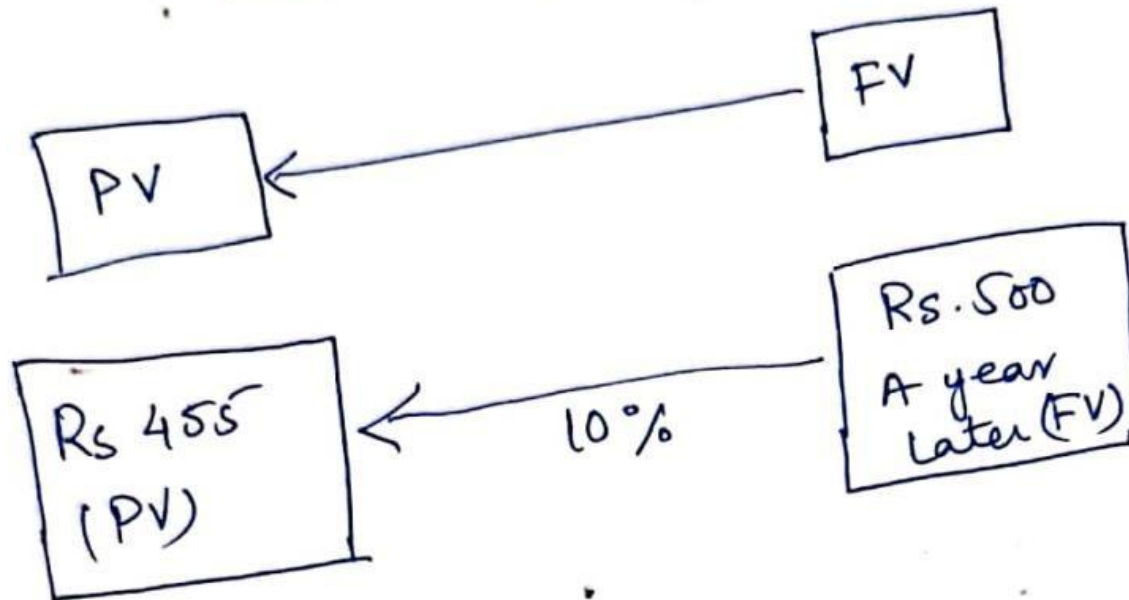
9.2 Numeric Methods/Evaluation Techniques (19/38)

B. Discounted techniques

↳ let's see what is discounting?

Discounting

PV = Present Value
FV = Future Value.



9.2 Numeric Methods/Evaluation Techniques (20/38)

B. Discounted techniques

Formula

$$PV = FV \cdot \frac{1}{(1+r)^n}$$

where $r = \text{interest rate} = 10\%$
 $n = \text{no. of years} = 1$

$$PV = 500 \times \frac{1}{(1 + 10\%)^1}$$

$$= 454.5$$

$$PV = 455 \text{ Rupees}$$

9.2 Numeric Methods/Evaluation Techniques (21/38)

B. Discounted techniques

FORMAT

Year	Cash Flows	Discounting Factor (DF)	Present Value (PV)
NPV			XXXXX

Decision Rule

- ✓ If $NPV > 0$; Accept the project
- ✓ If $NPV < 0$; Reject the project

9.2 Numeric Methods/Evaluation Techniques (22/38)

B. Discounted techniques

Example 5

Calculate NPV of two projects and suggest which of the two projects should be selected? Assuming discount rate @ 10%.

	Project X	Project Y
Initial Investment	Rs. 40,000	Rs. 60,000
Estimated Life	5 years	5 years
Scrap Value	Rs. 2,000	Rs. 4,000

9.2 Numeric Methods/Evaluation Techniques (23/38)

B. Discounted techniques

Cash Inflows					
Year	1	2	3	4	5
Project X	10,000	20,000	20,000	6,000	4,000
Project Y	40,000	20,000	10,000	6,000	4,000

PV Factor @ 10% Discount					
Year	1	2	3	4	5
PV Factors @ 10%	0.909	0.826	0.751	0.683	0.621

9.2 Numeric Methods/Evaluation Techniques (24/38)

B. Discounted techniques

Year	Cash flows	DF@10%	PV
0	(40,000) → PV/ Cash outflow	1	(40,000)
1	10,000 → FV	$0.909 \rightarrow (1 \div 1.10)^1$	$= FV \times DF@10\%$ $= 10,000 \times 0.909 = 9,090 \text{ rps}$
2	20,000 → FV	$0.826 \rightarrow (1 \div 1.10)^2$	$= 20,000 \times 0.826$ $= 16,520 \text{ rps}$
3	20,000	0.751	15,020 rps
4	6,000	0.683	4,098 rps
5	4,000	0.621	2,484 rps
5 (Scrap)	2,000	0.621	1,242
		NPV _X	8,454 rupees

Project X

→ The value of 20,000 after a year would be 16,520 rps

9.2 Numeric Methods/Evaluation Techniques (25/38)

B. Discounted techniques

Discount rate 10%

Net present value for project X can be calculated as

$$NPV_X = PV_0 + PV_1 + PV_2 + PV_3 + PV_4 + PV_5 + PV_5(\text{scrap})$$

year 0 year 1

$$= -40000 + 9090 + 16520 + 15020 + 4098 + 2484 + 1242$$

$NPV_X = 8454 \text{ rupees}$

9.2 Numeric Methods/Evaluation Techniques (26/38)

B. Discounted techniques

Now let's evaluate Project Y.

Project Y

Year	Cash flows	DF@ 10%	PV
0	(60,000) → FV	1	(60,000)
1	40,000 → PV	0.909	$= FV \times DF@ 10\%$ $= 40,000 \times 0.909 = 3,636 \text{ rps}$
2	20,000	0.826	16520
3	10,000	0.751	7510
4	6,000	0.683	4098
5	4,000	0.621	2484
5 (scrap)	4,000	0.621	2484

9.2 Numeric Methods/Evaluation Techniques (27/38)

B. Discounted techniques

NPV_Y can be calculated as:

$$\begin{aligned} \text{NPV}_Y &= \text{PV}_0 + \text{PV}_1 + \text{PV}_2 + \text{PV}_3 + \text{PV}_4 + \text{PV}_5 + \text{PV}_{5(\text{scrap})} \\ &= -60,000 + 36,360 + 16,520 + 7,510 + 4,098 + 2,484 + 2,484 \end{aligned}$$

$$\boxed{\text{NPV}_Y = 9,456}$$

* The project with highest NPV will be selected, i.e.,
Project Y will be selected

9.2 Numeric Methods/Evaluation Techniques (28/38)

B. Discounted techniques

ii. Internal Rate of Return

↳ The IRR represents the discount rate at which the NPV of an investment is ZERO.

↳ It is a discounted cash-flow technique which takes into account the time value of money

↳ $PV \text{ of the expected cash inflows} = \text{Initial cash outflow}$

↳ IRR Acceptance Rule

✓ Accept if $IRR > K$

✓ Reject if $IRR < K$

$K = \text{cost of capital}$

↳ If these two factors are equal then only NPV is ZERO

9.2 Numeric Methods/Evaluation Techniques (29/38)

B. Discounted techniques

ii. Internal Rate of Return

Calculation

- 1) Calculate two NPVs (calculating two NPVs with two different assumed rates) for the project at two different costs of capital.

- 2) Use the following ^{formula} ↑ to find IRR

$$IRR = L + \underbrace{\left[\underbrace{\frac{N_L}{N_L - N_H}}_{2^{nd}} \times \underbrace{(H - L)}_{1^{st}} \right]}_{3^{rd}}$$

where L = lower Rate; H = High Rate; N_L = NPV at lower rate
 N_H = NPV at higher rate

9.2 Numeric Methods/Evaluation Techniques (30/38)

B. Discounted techniques

ii. Internal Rate of Return

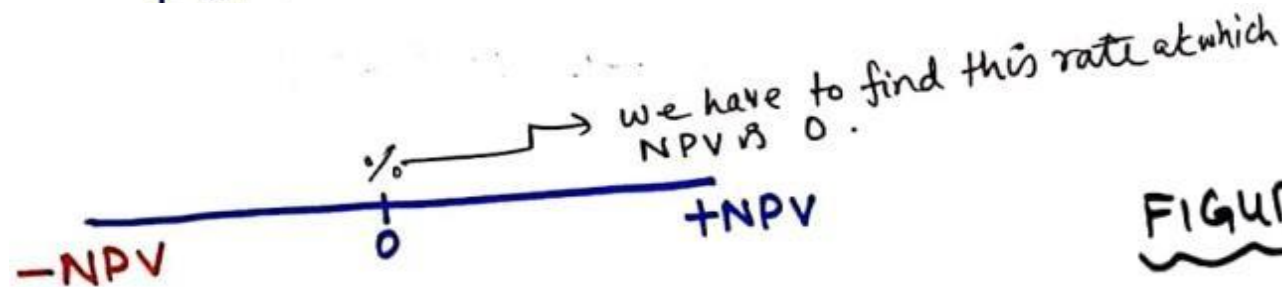
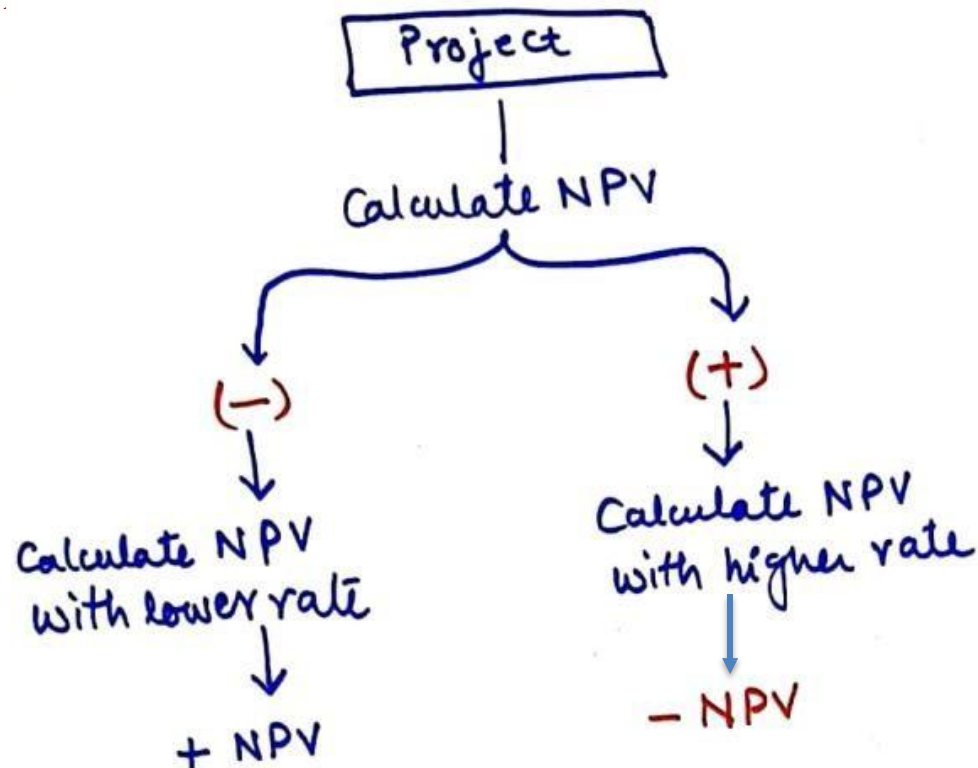


FIGURE 2

9.2 Numeric Methods/Evaluation Techniques (31/38)

B. Discounted techniques

ii. Internal Rate of Return

Example 6

Calculate the IRR of an investment of Rs. 1,36,000 which yield the following cash inflows.

Year	Cash Inflows
1	30,000
2	40,000
3	60,000
4	30,000
5	20,000

9.2 Numeric Methods/Evaluation Techniques (32/38)

B. Discounted techniques

ii. Internal Rate of Return

Solution

↳ In order to calculate IRR, first we need two NPVs, one +ve and one -ve and then we are going to use formula of IRR.

↳ Calculation of 1st NPV

Simply we have to discount all cash inflows given in table, Now to discount we need a discounting rate, In question the discounting rate is not given. We can assume any random discounting rate, Let's assume DF as 10%.

9.2 Numeric Methods/Evaluation Techniques (33/38)

B. Discounted techniques

ii. Internal Rate of Return

Year	Cash Inflows	DF@ 10%	PV	DF@ 12%	PV
0	(1,36,000)	1	(1,36,000)	1	(1,36,000)
1	30,000	$(1 \div 1.10)^1 = 0.909$	$30000 \times 0.909 = 27,270$	$(1 \div 1.12)^1 = 0.893$	$30000 \times 0.893 = 26,790$
2	40,000	$(1 \div 1.10)^2 = 0.826$	$40000 \times 0.826 = 33040$	$(1 \div 1.12)^2 = 0.797$	$40000 \times 0.797 = 31880$
3	60,000	0.751	45060	0.712	42720
4	30,000	0.683	20,490	0.636	19080
5	20,000	0.621	12,420	0.567	11340
		NPV(+)	2280	NPV(-)	-4910

9.2 Numeric Methods/Evaluation Techniques (34/38)

B. Discounted techniques

ii. Internal Rate of Return

↳ If first NPV is +ve using random DF (in this case DF@10%), now we need a -ve NPV (Refer to figure no 2).

↳ For -ve NPV we have to use higher Discounting rate i.e we assumed DF as 12% in above table related to example 6.

↳ Now we can apply the formula to interpolate

$$IRR = L + \left[\frac{N_L}{N_L - N_H} \times (H - L) \right]$$

$$\checkmark H - L = 12 - 10 = 2$$

$$\checkmark \frac{N_L}{N_L - N_H} = \frac{2280}{2280 + 4910} = 0.317$$

9.2 Numeric Methods/Evaluation Techniques (35/38)

B. Discounted techniques

iii. Internal Rate of Return

Now,

$$IRR = 10 + [0.317 \times 2]$$

$$\boxed{IRR = 10.63}$$

↳ If we use this rate i.e., DF@ 10.63% in our calculations our cash inflows (from year 1 to year 5) will be approximately equal to our initial investment. i.e.

✓ DF@ 10.63% gives total cash ⁱⁿflow of 136120, which is very much near to the initial investment amount i.e. 136000

9.2 Numeric Methods/Evaluation Techniques (36/38)

B. Discounted techniques

* Profitability Index (PI)

↳ The PI measures the ratio b/w the PV of future cash inflows and the PV of cash outflows

$$PI = \frac{\text{PV of cash inflows}}{\text{PV of cash outflows}}$$

↳ Acceptance Rule

✓ If $PI > 1$; Accept the project → NPV +ve

✓ If $PI < 1$; Reject the project → NPV -ve

✓ If $PI = 0$; NPV = 0 → we may accept/reject the project

↳ How to interpret the value of PI?

Say we have $PI = 1.2$ (in terms of dollars) ; we say that dollar 1.2 is generated for each dollar (1 \$) you invest.

9.2 Numeric Methods/Evaluation Techniques (37/38)

B. Discounted techniques

Example 7

Suppose we have three projects involving discounted cash outflow of Rs. 5,50,000, Rs. 75,000 and Rs. 1,00,20,000 respectively. Suppose further that the sum of discounted cash inflows for these projects are Rs. 6,50,000, Rs. 95,000 and Rs. 1,00,30,000 respectively. Calculate the desirability factors for the three projects.

9.2 Numeric Methods/Evaluation Techniques (38/38)

B. Discounted techniques

Solution

The desirability factors for three projects would be as follows:

$$1) \frac{\text{Rs. } 6,50,000}{\text{Rs. } 5,50,000} = 1.18$$

$$2) \frac{\text{Rs. } 95,000}{\text{Rs. } 75,000} = 1.27$$

$$3) \frac{\text{Rs. } 1,00,30,000}{\text{Rs. } 1,00,20,000} = 1.001$$

Reference

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